

11_API_Architecture.md

Part 1 — API Foundation

Document Information

Property	Value
Project	Aetheris Trade
Document	API Architecture
Version	1.0.0
Status	Draft
Owner	Priyanshu Rajpurohit

Purpose

This document defines the complete API architecture of Aetheris Trade.

It establishes:

- API standards
- Endpoint structure
- Authentication
- Versioning
- Request/Response conventions
- Security guidelines

This document is the single source of truth for backend API development.

API Philosophy

Every API should be:

- RESTful
- Predictable
- Secure
- Versioned
- Consistent
- Scalable

- Well documented
-

API Design Principles

- Resource-based URLs
 - Stateless requests
 - Consistent naming
 - Standard HTTP methods
 - Standard HTTP status codes
 - JSON responses only
-

Base URL

Development

```
```text id="0x7kt5" http://localhost:3000/api/v1
```

Production

```
```text id="w9kkm2"
https://api.aetheristrade.com/v1
```

API Versioning

Current Version

```
```text id="g65u0d" v1
```

Future

```
```text id="n6yppg"
v2
v3
```

Versioning Rule

```
```text id="6n7zq7" /api/v1/
```

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# Authentication

Provider

- Clerk

Method

- Bearer Token

Header

```text id="wth81v"

Authorization: Bearer <token>

HTTP Methods

| Method | Purpose |
|--------|------------------|
| GET | Read Data |
| POST | Create Data |
| PUT | Replace Resource |
| PATCH | Update Resource |
| DELETE | Delete Resource |

HTTP Status Codes

| Code | Meaning |
|------|--------------|
| 200 | Success |
| 201 | Created |
| 204 | No Content |
| 400 | Bad Request |
| 401 | Unauthorized |
| 403 | Forbidden |
| 404 | Not Found |

| Code | Meaning |
|------|-----------------------|
| 409 | Conflict |
| 422 | Validation Error |
| 429 | Too Many Requests |
| 500 | Internal Server Error |

API Naming Convention

Resources

``text id="ezqeqn" /users

/stocks

/portfolio

/watchlist

/transactions

Plural nouns only.

Request Format

```
```json
{
 "data": {}
}
```

---

## Success Response

```
{
 "success": true,
 "data": {},
 "message": "Request completed successfully."
}
```

---

## Error Response

```
{
 "success": false,
 "error": {
 "code": "VALIDATION_ERROR",
 "message": "Invalid request."
 }
}
```

## Pagination Standard

Query Parameters

``text id="i71xmg" ?page=1

?limit=20

Response

```
`` `json
{
 "data": [],
 "pagination": {
 "page": 1,
 "limit": 20,
 "total": 100,
 "totalPages": 5
 }
}
```

## Filtering Standard

Examples

``text id="9y2q2u" /stocks?sector=technology

/stocks?exchange=NSE

/transactions?type=BUY

---

# Sorting Standard

Example

```
``text id="nv9y9d"
?sort=created_at

?order=desc
```

---

## Search Standard

Example

```
text id="gjmjlwm"
/stocks?search=reliance
```

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## API Modules

- Authentication
- Users
- Portfolio
- Holdings
- Transactions
- Stocks
- Market Data
- Watchlist
- AI Coach
- Learning
- Competitions
- Leaderboard
- Notifications
- Admin

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## Documentation Standard

Every endpoint must include:

- Purpose
- Method
- URL
- Parameters

- Request Body
  - Success Response
  - Error Response
  - Authentication Requirement
- 

## Related Documents

- [10\\_Database\\_Architecture.md](#)
  - [12\\_Engineering\\_Standards.md](#)
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## Approval Status

Version: 1.0.0

Status: Draft

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## Next Document

[11\\_API\\_Architecture.md](#) — Part 2